
ASK ALPHA

The A–Z of Ask Alpha

Every use case the assistant supports, end to end — what triggers it, and how Ask Alpha responds.

AI assistant for UAE & GCC real estate — projects, developers, market intelligence, investment analysis, finance, documents, amenities and AI promo video.

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How to read this report

Ask Alpha is a tool-using AI agent. Each message you send is handled by a reasoning model that decides which of its specialist tools to call, runs them against the Allegiance database, and writes a grounded reply. This report walks through **26 use cases, lettered A to Z** — the 19 specialist tools plus the conversation behaviours and channels you will actually experience. Reference sections at the end cover tool routing, guardrails, channels and automation.

Each entry follows the same shape:

- **You say** — example phrasings that trigger the use case.
- **Under the hood** — which tool fires and what it does.
- **Ask Alpha replies** — a representative answer in Alpha’s voice.
- **If there’s no data / edge case** — what happens when the data is missing or access is denied.

IMPORTANT — ABOUT THE SAMPLE REPLIES

The numbers, project names and developers inside the example replies are **illustrative placeholders** written for this report. The real Ask Alpha never invents figures: every factual answer it gives is pulled from a tool result against the Allegiance database, and if the data isn’t there it says so. The wording of the “no data” and refusal replies, however, is quoted from the actual system.

The 26 use cases at a glance

A	search_projects	Find projects by name, location, budget or sale status
B	search_units	Find projects by unit attributes (beds, type, size, unit price)
C	get_project_details	Full detail on one project: pricing, units, amenities, brochure
D	search_nearby_projects	Projects near a place or coordinate, by distance
E	get_market_intelligence	Real transaction-based market data for an area
F	analyze_investment	Is this project a good investment? vs-market verdict
G	compare_projects	Compare 2–3 projects head-to-head
H	get_developer_profile	A developer's track record and reliability
I	calculate_mortgage	UAE mortgage: LTV cap, deposit, monthly payment
J	calculate_rental_yield	Gross and net rental yield
K	payment_plan_breakdown	Turn a payment plan (e.g. 60/40) into cash amounts
L	total_cost_of_ownership	All-in purchase cost incl. DLD, commission, fees
M	check_golden_visa	Golden Visa eligibility by property value
N	search_documents	Semantic search of brochure text
O	agentic_search	Deep, multi-facet brochure search with citations
P	get_nearby_amenities	Schools, hospitals, metro, malls near a project
Q	get_property_valuation	Property Monitor AVM valuation (when enabled)
R	create_promo_video	Generate an AI-avatar promo video (agents only)
S	check_my_video_status	Check / fetch the link for a requested video
T	Pagination	"Show more" — the next page of results
U	Multi-turn follow-ups	"What about the second one?" uses prior context
V	Out-of-scope guard	Refusing to answer from outside the database
W	Concept explanations	Defining how mortgages, escrow, Golden Visa work
X	Telegram onboarding	Verify an agent by phone before any chat
Y	Telegram logout	Unlink a Telegram account
Z	Proactive video push	Auto-notify on Telegram when a video is ready

Search & discovery

A search_projects — find projects by name, location, budget, status

The front door. Anytime a user names a project or asks to find projects matching project-level criteria (location, budget, sale status), this runs.

You say: “Show me Emaar projects in Dubai Marina” · “Any projects under 1M AED?” · “What’s on sale in Abu Dhabi?” · “Do you have Beachgate?”

Under the hood: Queries the published-projects table by name/description, location (city, region, district, country), sale status and a price band. Budgets are converted to numbers (“1M”->1,000,000) and rows with no/zero price are dropped. Returns up to 5, highest price first when a budget is set.

ASK ALPHA REPLIES

Here are 5 projects under AED 1,000,000:

- **Marina Vista** — Emaar, Dubai Marina — AED 950,000–1,000,000
- **Creek Rise** — Emaar, Dubai Creek Harbour — AED 880,000–990,000
- (...3 more...)

There are more matches in this range — want me to show the next 5?

IF THERE'S NO DATA / EDGE CASE

If nothing matches a named project, Alpha runs a fuzzy search and uses one of two exact lines.

With near-matches: “We don’t have Beachgate in our system yet. Here are similar projects we do carry:” then 2–3 suggestions with developer, city and price range.

With nothing similar: “We don’t have Beachgate in our system yet, and I don’t see anything similar.” It will never claim a project “might be listed under another name.”

B search_units — find projects by unit attributes

Used whenever the user mentions any UNIT-level attribute — bedrooms, unit type, size in sqft, or a per-unit price. search_projects can’t filter these, so Alpha is told to switch to this tool.

You say: “4-bed villa or townhouse under 10M” · “2BR apartments between 1M and 2M in Dubai Marina” · “Penthouses over 2,000 sqft” · “Studios under 600k”

Under the hood: Joins the units table, filters at the unit level (synonyms map “flat”->apartments, “villas”->villa, etc.), then rolls matching units up per project with a *matched_units* block (count, price range, bed range, sizes).

ASK ALPHA REPLIES

Found 3 projects with 4-bedroom villas or townhouses under AED 10M:

- **Cavalli Estates** — DAMAC, DAMAC Hills — 4BR villas from AED 8.6M (6 matching units)
- **The Acres** — Meraas, Dubailand — 4BR townhouses AED 7.9M–9.8M (11 matching units)
- (...1 more...)

IF THERE'S NO DATA / EDGE CASE

If no unit matches the filters, Alpha reports that plainly rather than widening silently — e.g. “I don’t see any 4-bed villas under AED 10M in our current inventory.” It can then offer to relax a constraint (budget or bedroom count) if you ask.

C get_project_details — the full picture of one project

The deep-dive after a search: full pricing, a units breakdown, amenities, payment terms, furnishing, service charge, and brochure / cover links.

You say: “Tell me more about that first one” · “Full details on project 642” · “What amenities and payment terms does Marina Vista have?”

Under the hood: Fetches one project by numeric ID and serialises the full record, including up to 30 units (type, beds, baths, size, price, layout) and brochure URL.

ASK ALPHA REPLIES

Marina Vista by Emaar — Dubai Marina

Status: On sale · Completion: Q4 2026 · Escrow: yes · Post-handover plan: yes

Price range: AED 950,000 – 3,200,000

Unit mix: 1–3BR apartments, 690–1,850 sqft

Amenities: infinity pool, gym, private beach access, kids' play area

Service charge: ~AED 18/sqft · Furnishing: unfurnished

Brochure: link provided.

IF THERE'S NO DATA / EDGE CASE

If the ID doesn't exist Alpha returns “No project found with id ...” internally and tells the user it can't find that project; it will offer to search by name instead.

D search_nearby_projects — proximity search

Distance-based discovery. Anchors on a named area (resolved to the centre of that area's projects) or explicit lat/lng, and returns projects sorted by distance.

You say: “Projects within 5km of Downtown” · “Developments near Palm Jumeirah” · “What's close to the marina?”

Under the hood: Computes great-circle (Haversine) distance in SQL. Radius defaults to 5km, max 25km. Each result carries a *distance_km*.

ASK ALPHA REPLIES

Projects within 5 km of Downtown Dubai, nearest first:

- **Burj Crown** — Emaar — 0.6 km
- **Act One | Act Two** — Emaar — 1.1 km
- **Downtown Views II** — Emaar — 1.4 km

Want me to widen the radius or filter by budget?

IF THERE'S NO DATA / EDGE CASE

If the place name can't be resolved to any project coordinates, the tool returns found=false and Alpha says it couldn't locate that area to search around — and asks you to name a nearby landmark or community instead.

Market intelligence & investment

E get_market_intelligence — real market data for an area

Transaction-grounded market read for a community, district or project — medians, momentum and activity from 680k+ recorded deals.

You say: “What are prices like in Business Bay?” · “Is JVC rising or cooling?” · “How's the Palm performing?”

Under the hood: Calls a database function that resolves the area and returns 12-month median price and rate (AED/sqm and /sqft), 90-day momentum vs the prior 90 days, transaction counts, off-plan share and an activity label (hot / healthy / cooling / quiet).

ASK ALPHA REPLIES

Business Bay — last 12 months

Median price: AED 1.62M · Median rate: AED 1,640/sqft

90-day momentum: rate up 3.8% vs the prior 90 days

Activity: healthy · Off-plan share: 41% · Deals (12m): 4,210

Read: prices are firming with steady volume.

IF THERE'S NO DATA / EDGE CASE

If there are no transactions for the area, the tool returns found=false and Alpha says: “We don’t have transaction data for that area in our system yet.” It will not estimate or borrow numbers from general knowledge.

F analyze_investment — is this a good buy?

The investment verdict for one project: its asking rate vs the area market, the premium/discount, momentum, supply, payment-plan signal and a clearly-labelled yield ESTIMATE — with any data gaps surfaced.

You say: “Is Marina Vista a good investment?” · “Is project 642 worth buying?” · “Good ROI on this one?”

Under the hood: Computes the project’s median AED/sqft from its units, compares to the district market median ($\pm 15\%$ bands -> above / in line with / below market), reads a yield band by unit type, and lists data_gaps.

ASK ALPHA REPLIES

Marina Vista — investment read

Asking rate: AED 1,780/sqft vs Dubai Marina market median AED 1,650/sqft -> **~8% premium** (in line with market).

Momentum: area rate up 2.9% (90d) · Supply: 410 units · Post-handover plan: yes.

Rental yield ESTIMATE: 5.5–6.5% gross (market-typical band by unit type — no observed rental data yet).

Note the estimate caveat before deciding.

IF THERE'S NO DATA / EDGE CASE

If the project isn’t found: “We don’t have that project in our system yet.” If it exists but has no priced units or no district market data, Alpha still answers but names the gap explicitly (e.g. “no market transaction data for this district”) rather than fabricating a verdict.

G compare_projects — head-to-head

Runs the same investment metrics across 2–3 projects so they can be weighed side by side.

You say: “Compare Marina Vista and Creek Rise” · “Which is the better buy — 642, 643 or 644?”

Under the hood: Analyses each project ID with the same engine as F and returns them together.

ASK ALPHA REPLIES

Comparing two projects:

Marina Vista — AED 1,780/sqft, ~8% premium, yield est. 5.5–6.5%, post-handover: yes

Creek Rise — AED 1,510/sqft, ~3% below market, yield est. 6–7%, post-handover: no

Creek Rise looks better on entry price and yield; Marina Vista offers the softer payment plan. Your call depends on cash-flow vs. financing.

IF THERE'S NO DATA / EDGE CASE

If none of the supplied IDs are found, Alpha says none of those projects were in the system. Any project with gaps carries its own data_gaps note in the comparison.

H get_developer_profile — track record & reliability

A developer's portfolio at a glance: how many projects, delivered vs upcoming, an on-time delivery rate where dates exist, geographic spread, price range and notable projects.

You say: "Tell me about DAMAC's track record" · "Is Sobha reliable?" · "What has Emaar built?"

Under the hood: Aggregates the developer's published projects into counts, an on-time ratio (completion vs construction-end date) and a top-5 notable list by price.

ASK ALPHA REPLIES

Emaar Properties

Total projects: 128 · On sale: 22 · Delivered: 96 · Upcoming: 10

On-time delivery: ~88% (of dated projects) · Active in 14 districts, 3 cities

Price range: AED 720k – 45M

Notable: Burj Khalifa district towers, Dubai Creek Harbour, Emaar Beachfront...

IF THERE'S NO DATA / EDGE CASE

If the developer isn't on file: "We don't have that developer in our system yet." Where no completion dates exist, the on-time figure is reported as "no delivery dates on record" rather than guessed.

Finance calculators

These five tools do transparent arithmetic on the numbers you provide — they do not read the database, and every assumption (LTV caps, fee percentages, thresholds) is stated back to you. If a required number is missing, Alpha asks for it.

I calculate_mortgage — UAE mortgage maths

Applies the correct UAE loan-to-value cap, then returns deposit, loan, monthly payment and total interest.

You say: "Monthly payment on a 2M apartment?" · "How much deposit do I need on a 6M villa as a non-resident?"

Under the hood: LTV caps: resident first home <5M -> 80%; first home >=5M -> 70%; not first home -> 65%; non-resident <5M -> 60%, >=5M -> 50%. Rate defaults 4.5%, term 25y. If the chosen deposit is below the legal minimum it's raised automatically.

ASK ALPHA REPLIES

AED 2,000,000 apartment, resident, first home

Max LTV 80% -> minimum 20% down = AED 400,000

Loan: AED 1,600,000 @ 4.5% over 25y

Monthly payment: ~AED 8,900 · Total interest: ~AED 1.07M

Assumptions: UAE LTV caps; rate/term are inputs, not advice.

IF THERE'S NO DATA / EDGE CASE

If price is missing, Alpha returns an error asking for it. If a user requests, say, 10% down on a first home, Alpha raises it to the 20% legal minimum and notes that it did so.

J calculate_rental_yield — gross & net yield

Turns a price and an annual rent into gross and net yield, deducting service charge and other costs for the net figure.

You say: “If it rents for 120k on a 1.8M flat, what’s my yield?” · “Net yield after 15k service charge?”

Under the hood: Gross = rent / price. Net = (rent - service charge - other costs) / price.

ASK ALPHA REPLIES

Price AED 1,800,000 · Rent AED 120,000/yr · Service charge AED 15,000

Gross yield: 6.67% · Net yield: 5.83%

Net annual income: AED 105,000.

IF THERE'S NO DATA / EDGE CASE

If price or annual rent is missing, Alpha asks for the missing number instead of guessing.

K payment_plan_breakdown — plan to cash amounts

Converts a developer payment plan (during-construction / on-handover / optional post-handover %) into AED amounts per milestone.

You say: “What would a 60/40 plan cost me on a 2.5M unit?” · “Break down 50/10/40 post-handover over 3 years”

Under the hood: Splits the price by the percentages and flags whether they sum to 100%.

ASK ALPHA REPLIES

AED 2,500,000 on a 60/40 plan

- During construction (60%): AED 1,500,000

- On handover (40%): AED 1,000,000

Total checks out to 100%.

IF THERE'S NO DATA / EDGE CASE

If the percentages don’t add up to 100%, Alpha shows the split but warns “Percentages don’t sum to 100% — check the plan.”

L total_cost_of_ownership — the all-in cost

The true cost of buying in Dubai: price plus DLD transfer (4%), agent commission (2%), admin, and — if mortgaged — mortgage registration.

You say: “What’s the real cost on top of a 3M price?” · “Closing costs if I mortgage 2.4M of it?”

Under the hood: DLD 4% + commission 2% + AED 4,200 admin + (0.25% of loan + AED 290 if mortgaged).

ASK ALPHA REPLIES

AED 3,000,000, mortgaged with a 2.4M loan

DLD transfer (4%): AED 120,000 · Agent commission (2%): AED 60,000

Admin: AED 4,200 · Mortgage registration: AED 6,290

Total upfront fees: AED 190,490 (6.35% of price)

All-in cost: AED 3,190,490.

IF THERE'S NO DATA / EDGE CASE

If price is missing, Alpha asks for it. All percentages are stated as assumptions so you can adjust them.

M check_golden_visa — residency by property

Checks a property value against the AED 2,000,000 Golden Visa (real-estate route) threshold and explains eligibility.

You say: “Does a 2.2M apartment get me the Golden Visa?” · “Am I short of the residency threshold at 1.7M?”

Under the hood: Compares value to AED 2M; computes the shortfall if below.

ASK ALPHA REPLIES

AED 2,200,000 property

Eligible: yes — at or above AED 2,000,000 this can qualify the buyer for a 10-year UAE Golden Visa (real-estate route). Off-plan from approved developers and mortgaged properties can also qualify, subject to conditions.

Final eligibility is set by ICP/GDRFA.

IF THERE'S NO DATA / EDGE CASE

Below threshold (e.g. AED 1.7M): Alpha reports “not eligible” and the exact shortfall (“below the AED 2,000,000 threshold by AED 300,000”).

Documents & brochures (RAG)

N search_documents — semantic brochure search

Searches the prose inside ingested marketing materials — payment plans, amenity detail, finishings, location narrative, ROI language — that doesn't live in structured columns.

You say: “What does the Catch Residence brochure say about amenities?” · “Payment terms after handover for project 642”

Under the hood: Embeds the query and runs a vector similarity search over brochure chunks (optionally scoped to one project), returning the closest passages with a similarity score.

ASK ALPHA REPLIES

From the Marina Vista brochure (3 relevant passages):

- “A 10% deposit secures the unit, with 50% across construction milestones and 40% on handover...”
- “Residents enjoy an infinity-edge pool, private beach access and a dedicated concierge...”

These are quoted from the brochure on file.

IF THERE'S NO DATA / EDGE CASE

If no brochure has been ingested for the project, the search returns zero chunks and Alpha says it doesn't have brochure material for that project yet — it won't paraphrase from general knowledge.

O agentic_search — deep, multi-facet document search

For broad or multi-part questions, Alpha decomposes the question into facets (payment terms, amenities, location, price/value), retrieves for each, then fuses the best passages with citations — higher recall than a single search.

You say: “Tell me everything about project 642's payment plan and amenities” · “Is this a good investment?”
(when answered from brochure prose)

Under the hood: Runs up to 5 sub-queries in parallel, dedupes by passage keeping the best match, and tags each result with the facet that found it.

ASK ALPHA REPLIES

Here's what the brochures cover, by theme:

Payment — “60/40 plan with a 2-year post-handover option...”

Amenities — “rooftop infinity pool, smart-home wiring, EV charging...”

Location — “7 minutes to Downtown, direct Sheikh Zayed Road access...”

Each line is cited to the source passage.

IF THERE'S NO DATA / EDGE CASE

Like search_documents, if there's no ingested material the fused result is empty and Alpha says so plainly. The facet expansion is deterministic, so results are stable.

Amenities & valuation

P get_nearby_amenities — what's around a project

Lists real amenities near a project — schools, hospitals, clinics, pharmacies, malls, supermarkets, metro, parks, beaches — grouped by category with distances.

You say: “What schools are near Marina Vista?” · “Is there a hospital close to project 642?” · “How far is the metro?”

Under the hood: Uses a cached OpenStreetMap dataset for the project (lazily fetched once via Overpass), computing distances from the project's coordinates. Optional category filter.

ASK ALPHA REPLIES

Near **Marina Vista** (within 3 km):

Schools — Emirates International School (1.2 km), Dubai British School (2.4 km)

Metro — DMCC Station (0.7 km)

Supermarket — Carrefour Marina Mall (0.5 km)

Source: OpenStreetMap.

IF THERE'S NO DATA / EDGE CASE

If the project has no coordinates on record, Alpha replies that it has no location for that project, so it can't map amenities. If a category simply has nothing nearby, that category is omitted.

Q get_property_valuation — Property Monitor AVM

An automated market valuation (AVM) for a specific property, to check an asking price against estimated value and observed yields.

You say: “What’s a 1,200 sqft 2BR in Dubai Marina really worth?” · “Is this asking price fair?”

Under the hood: Resolves the area to a Property Monitor location and requests an AVM report for the size / bedrooms / property type.

ASK ALPHA REPLIES

Valuation — 2BR, 1,200 sqft, Dubai Marina

Estimated market value: AED 2.35M (range 2.20–2.50M)

An asking price within that band is fair; above it is a premium.

(A full report with comparables may require an OTP to unlock.)

IF THERE'S NO DATA / EDGE CASE

Current state: until the Property Monitor API keys are configured, this tool returns configured=false, and Alpha tells the user: “Property Monitor valuations aren’t enabled yet... once the keys are configured I can value a property against the market and report observed rental yields.” If the area isn’t found, it says so.

AI promo video (agents only)

R create_promo_video — generate an AI-avatar video

Generates a short vertical (Reels/TikTok) promo video about a project using a HeyGen AI avatar — with an optional custom script and AI-generated background scene.

You say: “Make a promo video for project 6” · “Create a marketing video for Damac Island in Zain’s voice” · “One for Rami on Monte Carlo with the Burj behind me”

Under the hood: Verifies the user is an agent, resolves the target person’s HeyGen avatar + voice (defaults to the requester; agent_name dispatches for a teammate), writes a Hook/Value/CTA script from the project’s data if none is given, optionally generates a background scene, and starts the render asynchronously. Multiple videos can be requested in one message.

ASK ALPHA REPLIES

Your video for **Damac Island** is being generated now — it usually takes 1–2 minutes. On Telegram I’ll send the download link automatically the moment it’s ready; on the website, just ask “is my video ready?” to check.

Format: 1080×1920 vertical · for: Zain.

IF THERE'S NO DATA / EDGE CASE

Not an agent / not signed in: “Sign in required — this feature is only for our agents.” **Avatar or voice missing in HeyGen** for the named person: Alpha returns a helpful error listing the avatars/voices that do exist, and asks you to create the missing one. If HeyGen is unreachable, it reports a video-service error.

S check_my_video_status — is my video ready?

Checks the status of a previously-requested video and hands back the shareable link when it’s done.

You say: “Is my video ready?” · “Where’s my video?” · “Send me the link” · “Status of video <id>”

Under the hood: Looks up the agent's video (a specific ID, or the most recent) and returns its status and, when completed, the video URL and thumbnail.

ASK ALPHA REPLIES

Completed: "Your video for Damac Island is ready — here's the link to download or share: <url>."

Still rendering: "It's still processing — try again in a minute."

IF THERE'S NO DATA / EDGE CASE

Failed render: Alpha reports the failure with the reason HeyGen gave. **No videos yet / not an agent:** it says there are no videos for you, or that the feature is for agents only. A user can only see their own videos.

Conversation behaviours

T Pagination — "show me more"

Results come 5 at a time. When more exist, Alpha offers the next page and remembers your filters when you accept.

You say: "Show more" · "Next 5" · "Yes, the rest"

Under the hood: When a search returns `has_more`, Alpha mentions more are available; on "yes" it re-runs the same search with the next offset.

ASK ALPHA REPLIES

First reply ends with: "There are more matches — want the next 5?"

On "yes": "Here are the next 5 under AED 1M:" ... continuing the same query and filters from where it left off.

IF THERE'S NO DATA / EDGE CASE

Alpha shows at most 5 projects per reply by design, to keep answers scannable. If there are no further results it says that's everything it has.

U Multi-turn follow-ups — context carries over

The conversation is stateful: follow-ups like "the second one" or "what about its amenities?" resolve against the previous turn.

You say: "What about the second one?" · "And its payment plan?" · "Compare those two"

Under the hood: The last several messages are replayed to the model each turn, so it can map "the second one" to a specific project from its previous reply and call the right tool.

ASK ALPHA REPLIES

After listing projects, you ask "details on the second one?" — Alpha replies: "Creek Rise by Emaar — here's the full breakdown..." having mapped "the second one" to that project without you repeating its name.

IF THERE'S NO DATA / EDGE CASE

Context is a rolling window of recent messages. On the Telegram bot a restart clears the in-memory conversation pointer, so very old threads may start fresh.

V Out-of-scope guard — it won't make things up

Ask Alpha is strictly grounded in the Allegiance database. For specific facts not in the system, it declines rather than answering from training data.

You say: “What’s the price of <a project we don’t carry>?” · “Is <random developer> any good?” · “What are rents in <an area with no data>?”

Under the hood: A hard system rule forbids using external/training knowledge for any specific project price, developer, unit count, completion date, amenities, market price or yield. Missing data -> an honest “we don’t have that”.

ASK ALPHA REPLIES

“We don’t have that information in our system yet.”

— and for an unknown project specifically: “We don’t have <name> in our system yet, and I don’t see anything similar.”

IF THERE'S NO DATA / EDGE CASE

This is the core trust guarantee: Alpha never guesses a price, invents a project, or claims something “might be listed under another name.” If a tool returns nothing, the answer is “we don’t have it” — even when a public answer might exist.

W Concept explanations — general knowledge, allowed

For how-things-work questions (not specific listings), Alpha may use general knowledge — defining terms and explaining UAE buying mechanics.

You say: “How does escrow work here?” · “What does post-handover mean?” · “Explain ROI vs rental yield” · “Is Dubai Marina on the coast?”

Under the hood: The system prompt permits general knowledge for concepts, definitions and non-specific geographic context — but not for any specific project’s facts or live market numbers.

ASK ALPHA REPLIES

“Post-handover means part of the price is paid in instalments after you receive the keys, easing cash-flow. In the UAE, escrow law requires off-plan buyer funds to sit in a regulated account released to the developer against construction milestones...”

IF THERE'S NO DATA / EDGE CASE

The line is firm: concepts and definitions are fair game; the moment a question needs a specific project’s price/developer or an area’s live numbers, Alpha switches back to tools and the data-scope guard applies.

Channels & onboarding

X Telegram onboarding — verify before you chat

On Telegram, the bot is agent-only and must verify who you are before answering any real-estate question.

You say: /start · tapping “Share my phone to verify” · typing a phone number · any message before being verified

Under the hood: The bot asks you to share your Telegram contact (the secure path) or type your number. It matches the phone to a profile and checks it has an agent role and Ask Alpha access, then links the chat.

ASK ALPHA REPLIES

On /start: "Welcome to Ask Alpha. Tap 'Share my phone to verify' below..."

On a successful match: "Verified. Welcome, Zain. Ask me anything — e.g. 'Show me Damac projects' or 'Make a promo video for project 6'."

IF THERE'S NO DATA / EDGE CASE

No matching profile: "I couldn't find an agent profile linked to that number — contact your admin to be added." **Found but no agent access:** "Your profile doesn't have agent access." **Someone else's contact:** it refuses and asks for your own. Typing a number works but is flagged as less secure than the share button.

Y Telegram logout — unlink this chat

Lets an agent disconnect their Telegram chat from their profile.

You say: /logout

Under the hood: Deletes the messaging link for this chat and clears the conversation pointer.

ASK ALPHA REPLIES

"Unlinked. Send /start to verify again."

IF THERE'S NO DATA / EDGE CASE

After logout, any further message is treated as a new, unverified visitor and routed back into the /start verification flow before Alpha will answer.

Z Proactive video push — you'll be told when it's ready

When a video you requested on Telegram finishes rendering, Ask Alpha pushes you the link automatically — no need to keep asking.

You say: (No action needed — this happens after you request a video on Telegram via use case R.)

Under the hood: A background worker polls render status every 10 seconds; on completion it messages the originating Telegram chat with the link (and notifies on failure).

ASK ALPHA REPLIES

Arrives unprompted in your Telegram: "Your video is ready — Damac Island. Download / share: <url>" (with a check-mark).

If it failed: "Your video for Damac Island failed: <reason>."

IF THERE'S NO DATA / EDGE CASE

The push only fires for videos requested through Telegram (it needs the chat ID). On the website you poll with use case S instead. Notifications are sent once, after the status is safely saved, so you won't get duplicates.

Reference 1 — How Ask Alpha picks a tool

A quick map from what you ask to what fires. When a question spans several, Alpha chains tools in one turn (e.g. search -> details -> amenities).

Names a project, or filters by location / budget / status	A search_projects
Mentions bedrooms, unit type, sqft, or per-unit price	B search_units
Wants the full breakdown of one known project	C get_project_details
"near", "close to", "within N km"	D search_nearby_projects
Area prices, trends, rising/cooling, activity	E get_market_intelligence
"Good investment?", "worth buying?", "good ROI?"	F analyze_investment
"Compare X and Y", "which is better?"	G compare_projects
A developer's reputation / portfolio / reliability	H get_developer_profile
Mortgage, deposit, monthly payment	I calculate_mortgage
Yield, return, net income from price + rent	J calculate_rental_yield
"What would a 60/40 plan cost?"	K payment_plan_breakdown
True/total cost, closing costs, fees on top	L total_cost_of_ownership
Golden Visa / residency by property value	M check_golden_visa
Brochure prose: amenities, finishings, payment narrative	N search_documents
Broad / multi-part brochure questions	O agentic_search
Schools, hospitals, metro, malls near a project	P get_nearby_amenities
"What's it really worth?" / fair price (AVM)	Q get_property_valuation
"Make a promo / marketing / AI video"	R create_promo_video
"Is my video ready?" / "send the link"	S check_my_video_status

Reference 2 — Guardrails & data scope

- **Database-only facts.** Every project name, price, developer, unit count, completion date and brochure detail must come from a tool result. Training data about UAE real estate is off-limits for specific facts.
- **Honest gaps.** No data -> "We don't have that in our system yet." No invented projects, prices, or "maybe it's under another name."
- **Labelled estimates.** Rental-yield figures from analyze_investment are explicitly marked ESTIMATE with their basis; data gaps are surfaced, not hidden.
- **Concepts allowed.** General knowledge is fine for definitions and how UAE mechanics work (escrow, mortgages, Golden Visa) and non-specific geography.
- **Agent-gated actions.** Promo video creation and status are limited to verified agents (sales-agent or admin role, with Ask Alpha access).
- **Bounded reasoning.** Alpha runs up to ~10 tool steps per turn; if it can't resolve a request it says so rather than looping.

Reference 3 — Where you can use Ask Alpha

Website / app (REST API). The chat endpoint powers the web experience, with supporting endpoints for history and lookups:

POST /v1/chat	Send a message; get the reply + rich cards (project lists, market, video, ...).
GET /v1/conversations	List chats for a user (or anonymous).
GET /v1/conversations/{id}	Fetch one conversation.
GET /v1/conversations/{id}/messages	Full message history of a conversation.
GET /v1/projects · /v1/projects/{id}	List / fetch project records directly.
GET /v1/videos/{id}	Fetch a video job's status and URL.
GET /health	Service health check.

Telegram bot. The same brain, agent-only, with phone verification (X), logout (Y), and proactive video delivery (Z). Replies arrive as plain text plus formatted summaries of any cards.

Reference 4 — Behind the scenes

- **Document ingestion.** Project brochures (PDFs in storage) are text-extracted — native text or OCR for scanned files — chunked, embedded, and stored as vectors that power use cases N and O.
- **Video poller.** A background loop checks in-flight video renders every 10 seconds, updates their status, and triggers the Telegram push (Z).
- **Amenity cache.** The first amenities question for a project fetches nearby places from OpenStreetMap once and caches them for fast repeat answers (P).
- **Rich cards.** Most tool results also return structured cards so the web UI can render project lists, market panels, investment analyses and video status visually — the same data the text reply is built from.

End of report — 26 use cases (A–Z) plus channels, guardrails and automation, generated from the Ask Alpha codebase.